CSCI 264 — Assembly Language

Lab 7

The Triangle Inequality

Three lengths a, b, and c can be the sides of a triangle only if each pair of sides adds up to more than the third side:

$$a + b > c \qquad b + c > a \qquad a + c > b$$

This is called the **triangle inequality**. A valid triangle must satisfy **all three** combinations.

Setup

Work in the ASM Visualizer's Arithmetic Mode, as in Lab 6:

`https://asm.diveintosystems.org/arithmetic/x86_64`

Save your code to a local .txt file as you go.

In both parts, the variables start on the stack. Set them in the **Initial Stack Values** boxes: `-8(%rbp)` is the `%rbp - 8` box, `-16(%rbp)` is the `%rbp - 16` box, and `-24(%rbp)` is the `%rbp - 24` box.

In every part:

- Put the result in `%rax`.
- Use `%rcx`, `%rdx`, and `%r8` through `%r11` for anything else you need.
- You may use any instruction we have covered, but you should only need `mov`, `add`, `cmp`, and jumps.
- You may assume the side lengths are positive.
- Put comments in your code to help with the demos.

Part 1: Is It a Triangle?

Write an Assembly program which checks three values against the triangle inequality. Put 1 in `%rax` if they make a triangle and 0 if they don't.

The side lengths should start on the stack: `a` is at `-8(%rbp)`, `b` is at `-16(%rbp)`, and `c` is at `-24(%rbp)`.

`cmp` can't compare two memory operands, so you will have to store some values in registers.

Test with these values. The first two give the answer you should get. For the rest, work out the answer by hand *before* you run it, then check it.

a	b	c	%rax
3	4	5	1
1	2	3	0
5	5	5	
1	1	5	
5	1	1	
1	5	1	

Part 2: Counting Triangles

Now you are given two fixed values, a and b, and a limit n. Write an Assembly program which counts how many whole numbers `c` from 1 to `n` make a triangle with `a` and `b`, and leave the count in `%rax`. Use the ASM visualizer, start with the values on the stack with `a` at `-8(%rbp)`, `b` at `-16(%rbp)`, and the limit `n` at `-24(%rbp)`.

Start from your Part 1 code.

Test with these values. The first one gives the answer you should get. For the rest, work out the answer by hand *before* you run it.

a	b	n	%rax
3	5	10	5
4	4	10	
3	5	4	
3	5	0	

Submission

Finish Part 1 before you leave lab today.

Submit `lab7.txt` on Canvas by Tuesday, November 3 and demo it to Professor West during Lab 8 that day. If a classmate helped you, put their name in the file. Disclose any use of Generative AI at the top.